

STAT 112

INTRODUCTION TO STATISTICS AND PROBABILITY II

SESSION 6- Geometric Random Variable & Distribution

Dr. Gabriel Kallah-Dagadu

Department of Statistics and Actuarial Science
University of Ghana

July 5, 2021

Session Outline

- 1 Geometric Random Variable
 - Geometric Experiment
- 2 Geometric Distribution
 - Moments
- 3 Computation of Probabilities

Session Overview

In this session, students are introduced to geometric random variable and its probability distribution. The experiment that gives rise to the geometric random variable and properties of the geometric distribution will be discussed. Computation of the probabilities using the geometric probability mass function and theoretical derivation of the mean and variance of the geometric random variable will be presented.

Learning Outcomes

At the end of the session, students will be able to:

- 1 Explain and describe the experiment leading to Geometric random variable.
- 2 State the probability mass function of the Geometric distribution and show that it is a probability function.
- 3 Prove the expectation and variance of Geometric distribution.
- 4 Compute probabilities and moments using the Geometric distribution function.

Geometric Random Variable

- There are three characteristics of a geometric experiment:
- There are one or more Bernoulli trials with all failures except the last one, which is a success.
- In theory, the number of trials could go on forever. There must be at least one trial.
- The probability, p , of a success and the probability, q , of a failure are the same for each trial. That is; $p + q = 1$ and $q = 1 - p$.

Geometric Experiment

- Let X denote the number of independent trials up to and including the occurrence of the first success.
- Each trial is Bernoulli with success probability p , $0 < p < 1$ and failure probability $q = 1 - p$.
- Let A be the event of the occurrence of a success, then $P(A) = p$ and $P(\bar{A}) = q$.
- Note that the experiment only terminates after the occurrence of the first success.
- Here X can assume values $1, 2, 3, \dots$

Geometric Experiment

- Suppose a success is achieved on the x^{th} trial, then we have this outcome:

Trial	1	2	3	...	$(x - 1)^{th}$	x^{th}
Outcome	$\bar{A}$	$\bar{A}$	$\bar{A}$	...	$\bar{A}$	A

- Note: We have $(x - 1)$ trials resulting in failures before the first success and the x^{th} trial result in a success.

Geometric Probability Mass Function

- The probability of success (x), is given as;

$$\begin{aligned}P(X = x) &= P(\bar{A}, \bar{A}, \dots, \bar{A}, A) \\&= P(\bar{A}) \times P(\bar{A}) \times \dots \times P(\bar{A}) \times P(A) \\&= [P(\bar{A})]^{x-1} \times P(A) \\&= q^{x-1} \times p.\end{aligned}$$

Geometric Probability Mass Function

The probability distribution of a Geometric random variable X , with parameter p , is given as:

$$P(X = x) = pq^{x-1}, \quad x = 1, 2, 3, \dots, \quad \text{and} \quad 0 < p < 1; \quad q = 1 - p.$$

The Geometric distribution with parameter p , is denoted as: $X \sim G(p)$.

Legitimacy of the PMF

Theorem 1:

If $X \sim G(p)$, then the its distribution mass function is a legitimate probability function.

Proof:

- 1 It is clear that $P(X = x) = pq^{x-1} > 0, \forall x$, since $0 < p < 1$.
- 2 $\sum_{all\ x} P(X = x) = 1$.

$$\begin{aligned}\Rightarrow \sum_{all\ x} P(X = x) &= \sum_{x=1}^{\infty} pq^{x-1} \\ &= p \sum_{x=1}^{\infty} q^{x-1} \\ &= p(1 + q + q^2 + q^3 + \dots)\end{aligned}$$

Proof of Legitimacy Conditions

Now, using the Geometric series for the sum of the first n terms with the first term been a and common ratio r , we have

$$S_n = \frac{a(1 - r^n)}{1 - r}, \text{ where } |r| < 1 \text{ or } S_n = \frac{a(r^n - 1)}{r - 1}, \text{ where } |r| > 1$$

Proof Cont'd:

The sum to infinity of the geometric series is:

$$S_\infty = \lim_{n \rightarrow \infty} S_n = \frac{a}{1 - r}, \text{ where } |r| < 1.$$

$$\begin{aligned} \sum_{x=1}^{\infty} P(X = x) &= p \sum_{x=1}^{\infty} q^{x-1} \\ &= p \frac{1}{1 - q} = p \frac{1}{p} = 1 \end{aligned}$$

Theorem 2:

If the random variable $X \sim G(p)$ the $E(X) = \frac{1}{p}$ and $Var(X) = \frac{q}{p^2}$.

Proof:

By definition,

$$\begin{aligned} E(X) &= \sum_{\text{all } x} xP(X = x) = \sum_{x=1}^{\infty} xpq^{x-1} \\ &= p \sum_{x=1}^{\infty} xq^{x-1} \end{aligned}$$

Note: Solving the expression $\sum_{x=1}^{\infty} xq^{x-1}$, can be done in two ways, thus using geometric series method or using differentiation method.

Proof Cont'd:

$$\text{let } U = \sum_{x=1}^{\infty} xq^{x-1} = 1 + 2q + 3q^2 + 4q^3 + \dots$$

This type of series is known as **Arithmetico-Geometric**, because its terms are partially Arithmetic progression (A.P.) and partially Geometric progression (G. P.).

Method I

$$U = 1 + 2q + 3q^2 + 4q^3 + \dots \quad (1)$$

$$qU = q + 2q^2 + 3q^3 + 4q^4 + \dots \quad (2)$$

Now eqn.(1)-eqn.(2), yield the result

$$U - qU = 1 + q + q^2 + q^3 + q^4 + \dots$$

$$\Rightarrow (1 - q)U = \sum_{r=0}^{\infty} q^r = \frac{1}{1 - q},$$

Proof Cont'd:

That is summation of G.P. to infinity with the first term, $a = 1$ and the common ratio, $r = q$.

Method I

$$\text{Now, } U = \frac{1}{(1-q)^2}$$

$$\therefore E(X) = p \sum_{x=1}^{\infty} xq^{x-1} = pU$$

$$= p \times \frac{1}{(1-q)^2} = p \times \frac{1}{p^2}$$

$$= \frac{1}{p}$$



Method II

Let $U = \sum_{x=1}^{\infty} xq^{x-1}$, and $\frac{d}{dq}q^x = xq^{x-1}$, we have

$$U = \sum_{x=1}^{\infty} \frac{d}{dq}q^x = \frac{d}{dq} \sum_{x=1}^{\infty} q^x,$$

$$U = \frac{d}{dq} \left(\frac{q}{1-q} \right)$$

$$U = \frac{(1-q) - q(-1)}{(1-q)^2} = \frac{1}{(1-q)^2} = \frac{1}{p^2}$$

$$\therefore E(X) = pU = p \times \frac{1}{p^2} = \frac{1}{p}$$

Note: Since the sum is uniformly convergent over $0 < q < 1$, we have the result by summing the G.P. series q^x , to infinity.

Proof Cont'd:

Variance:

By definition, the Variance of a random variable X is given as

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = E(X^2) - \frac{1}{p^2}$$

$$\begin{aligned} E(X^2) &= \sum_{\text{all } x} x^2 P(X = x) = \sum_{x=1}^{\infty} x^2 \cdot pq^{x-1} \\ &= \sum_{x=1}^{\infty} [x(x-1) + x] \cdot pq^{x-1} \\ &= \sum_{x=1}^{\infty} x(x-1) \cdot pq^{x-1} + \sum_{x=1}^{\infty} x \cdot pq^{x-1} \\ &= \sum_{x=1}^{\infty} x(x-1) \cdot pq^{x-1} + E(x) \end{aligned}$$

Proof of Variance:

$$E(X^2) = \sum_{x=1}^{\infty} x(x-1) \cdot pq q^{x-2} + E(x)$$

$$= pq \sum_{x=1}^{\infty} x(x-1) \cdot q^{x-2} + E(x)$$

$$= pq \sum_{x=1}^{\infty} \frac{d^2}{d q^2} q^x + E(x)$$

$$= pq \frac{d^2}{d q^2} \sum_{x=1}^{\infty} q^x + E(x)$$

$$= pq \frac{d^2}{d q^2} \left(\frac{q}{1-q} \right) + \frac{1}{p}$$

$$= pq \frac{d}{d q} \left\{ \frac{d}{d q} \left(\frac{q}{1-q} \right) \right\} + \frac{1}{p}$$

Proof of Variance:

$$\begin{aligned}E(X^2) &= pq \frac{d}{dq} \left\{ \frac{1}{(1-q)^2} \right\} + \frac{1}{p} \\&= pq \frac{2}{(1-q)^3} + \frac{1}{p} = \frac{2pq}{(p)^3} + \frac{1}{p} \\&= \frac{2q}{p^2} + \frac{1}{p}\end{aligned}$$

$$\begin{aligned}\therefore \text{Var}(X) &= \frac{2q}{p^2} + \frac{1}{p} - \frac{1}{p^2} \\&= \frac{2q + p - 1}{p^2} = \frac{2q - (1 - p)}{p^2} = \frac{2q - q}{p^2}\end{aligned}$$

$$\therefore \text{Var}(X) = \frac{q}{p^2}$$



Example 1

Suppose an oil-prospecting firm will drill in successions of holes in a given offshore area to find a productive well. The probability of a success on a given trial is 0.3. The cost of drilling a hole is GHc 10,000.00. If a hole is drilled and does not yield a productive well, an additional cost of GHc 3000.00 is increased because certain changes that must be made before the next drilled is attempted.

- a. If the drilling of holes continue until the first successful result is achieved, what is the expected cost of the entire operation?
- b. Find the probability that at most the third drilled is the first that yield a productive well.
- c. If the company has enough resources to sustain the drilling up to ten holes, determine the expected cost of the operation.

Solution:

Let X denote the number of drills up to and including the first successful drill.

$\implies X \sim G(p)$, where $p = 0.3$, the success probability.

$\therefore P(X = x) = 0.3(0.7)^{x-1}$, where $x = 1, 2, 3, \dots$

a:

Let the function $C(x)$ denote the cost of the entire operation, then

$$C(x) = 10,000X + 3,000(X - 1) = 13,000x - 3,000$$

$$\therefore E[C(x)] = 13,000E(x) - 3,000$$

$$E[C(x)] = 13,000\left(\frac{1}{0.3}\right) - 3,000 = 40,333.33$$

Since the $E(X) = 1/p$

The probability that at most the third drill that yield the productive well is:

b:

$$\begin{aligned}P(X \leq 3) &= \sum_{x=1}^3 0.3(0.7)^{x-1} \\&= 0.3(1 + 0.7 + 0.7^2) \\&= 0.3(1 + 0.7 + 0.49) \\&= 0.657\end{aligned}$$

Solution Cont'd:

If the company has enough resources to sustain the drilling up to ten holes, then the process is no more a geometric but a binomial process with a fixed number of trials. The expected cost for drilling ten holes is:

C:

$$\therefore E[C(x)] = 13,000E(x) - 3,000$$

where the random variable $X \sim b(10, 0.3)$

$$E[C(x)] = 13,000(10(0.3)) - 3,000 = 36,000$$

Since the $E(X) = np$.

Trial Questions:

- 1 Suppose Komla play a game of chance that he can either win or lose (there are no other possibilities) until he lose. His probability of losing is $p = 0.57$. What is the probability that it takes five games until Komla lose?
- 2 Gabby throw darts at a board until he hit the centre area. His probability of hitting the centre area is $p = 0.17$. Gabby is interested to find the probability that it takes eight throws until he hit the centre. Also, what values does X take on?
- 3 An instructor feels that 15% of students get below a C on their final exam. She decides to look at final exams (selected randomly and replaced in the pile after reading) until she finds one that shows a grade below a C. What is the probability that the instructor will have to examine at least ten exams until she finds one with a grade below a C?

Trial Questions:

- 4 A safety engineer feels that 35% of all industrial accidents in her plant are caused by failure of employees to follow instructions. She decides to look at the accident reports (selected randomly and replaced in the pile after reading) until she finds one that shows an accident caused by failure of employees to follow instructions.
- a. On average, how many reports would the safety engineer expect to look at until she finds a report showing an accident caused by employee failure to follow instructions?
 - b. What is the probability that the safety engineer will have to examine at least three reports until she finds a report showing an accident caused by employee failure to follow instructions?

Trial Questions:

- 5 Suppose that the probability of a defective computer component is 0.02. Components are randomly selected.
- a. Find the probability that the first defect is caused by the seventh component tested.
 - b. How many components do you expect to test until one is found to be defective?
- 6 Find the probability that in successive tosses of a fair die, a 3 will come up for the first time on the fifth toss.

THANK YOU!!!